The Impact of Weather on Flight Delays

Capstone project

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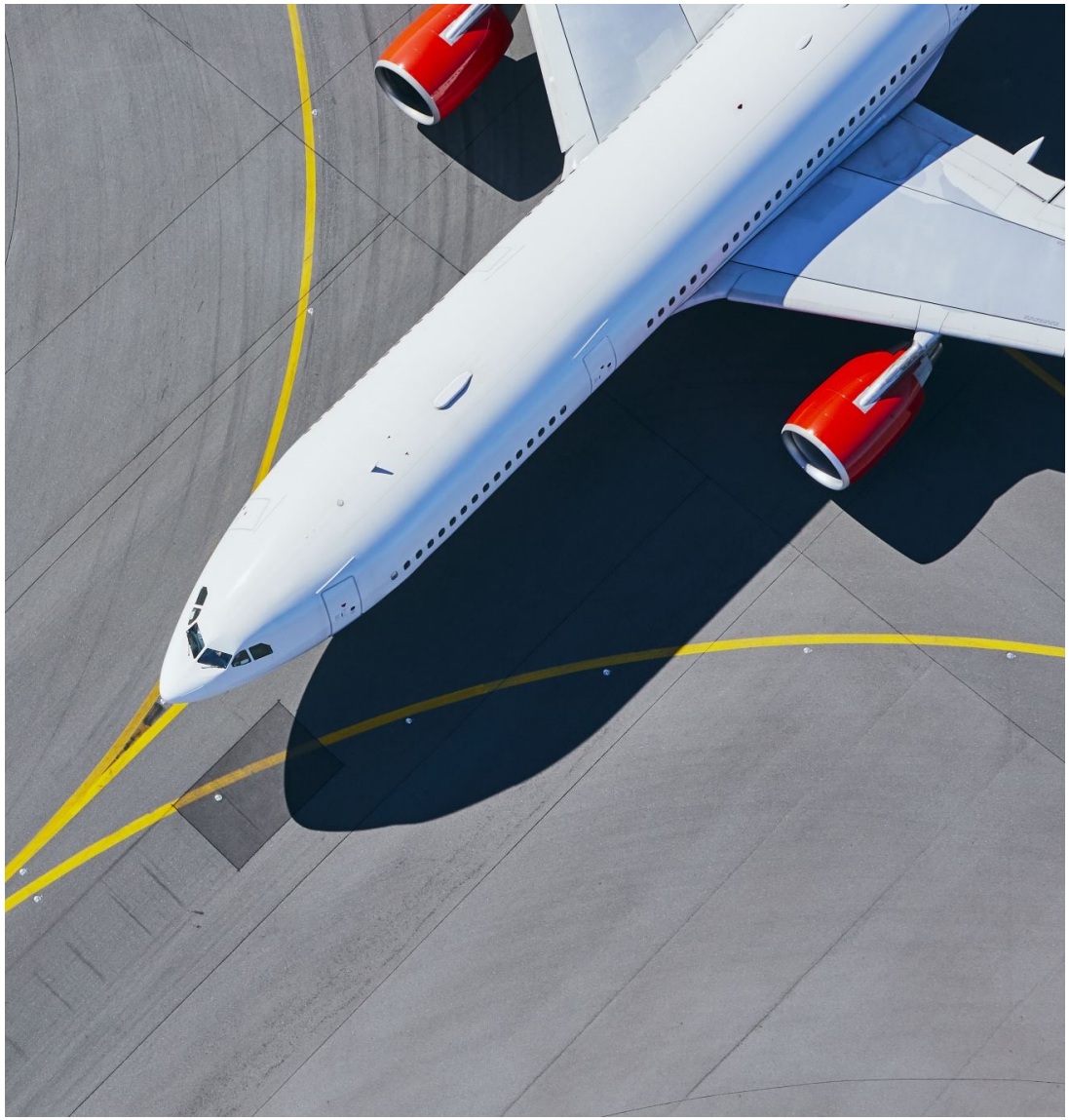
Agenda

- Business application
- Dataset
- Data exploration, analysis and visualisation
- Models used
- Results evaluation
- Summary, conclusions and next steps
- Any questions?

Business application

- Flight delays are not just an inconvenience — they represent a multi-billion-dollar operational challenge for the aviation industry. In the U.S. alone, delays have been estimated to cost around $33 billion yearly.
- The objective of this project is to examine the extent to which weather conditions influence the likelihood of flight departure delays and to develop a predictive model using combined flight and meteorological data.
- Such a model could support operational stakeholders across the aviation ecosystem — including air traffic management, airlines, and ground operations — enabling earlier planning and helping minimise the downstream impact of delays.



Dataset

Individual csv files downloaded from the United States Department of Transportation website, Bureau of Transportation Statistics, ‘On-Time: Reporting Carrier On-Time Performance (1987-present)’, with certain fields

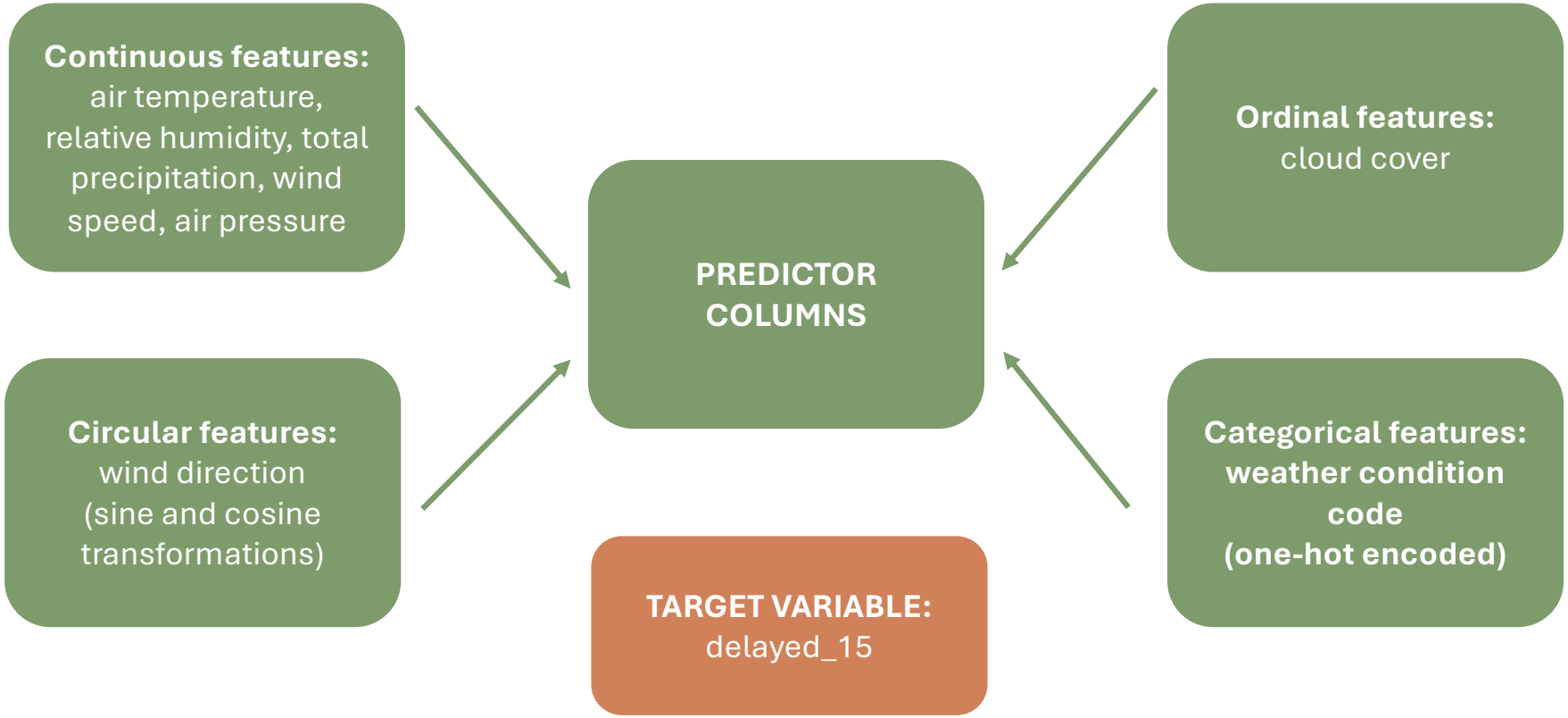
Train set are 2022 - 2023 flight data, limited to John F. Kennedy International Airport (JFK). 261,001 observations with 33 columns (not all used)

Test set are January – October 2025 flight data, limited to JFK. 85,435 observations with 33 columns (not all used)

Weather conditions at the scheduled departure time were added using Meteostat Python library

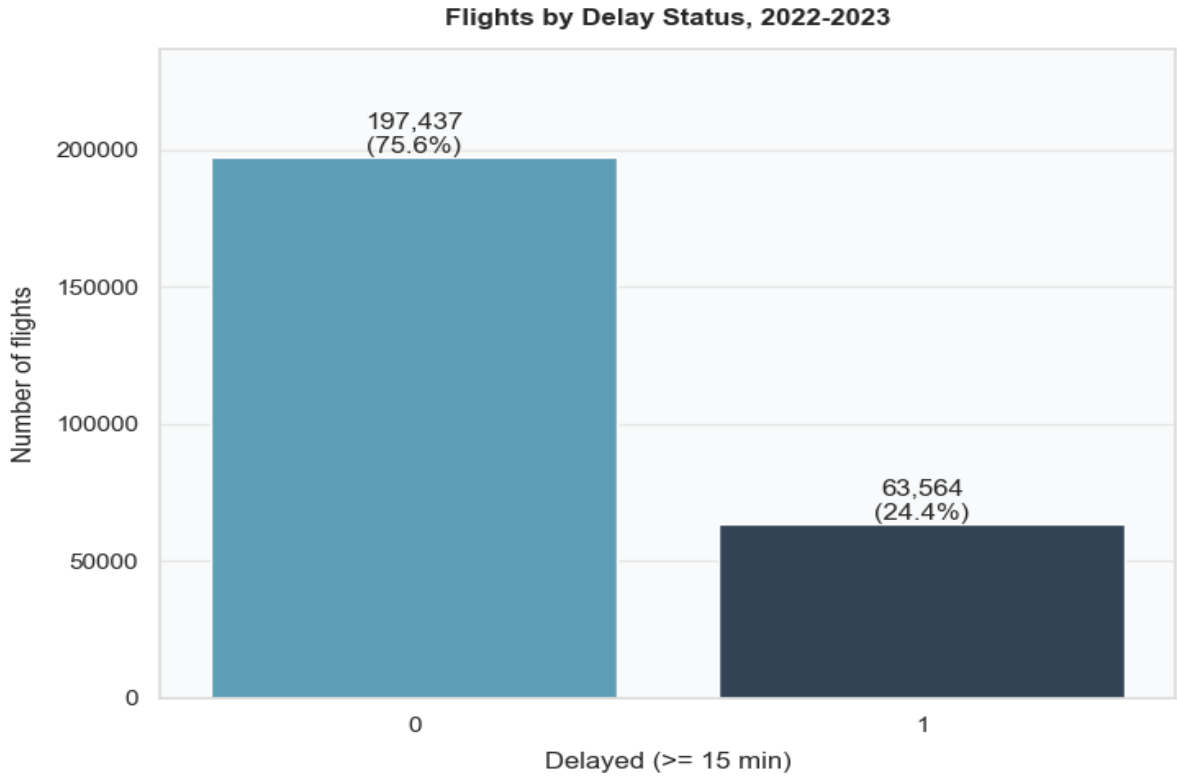
Delay is classified as departure from the gate 15 or more mins later than scheduled

Features and target



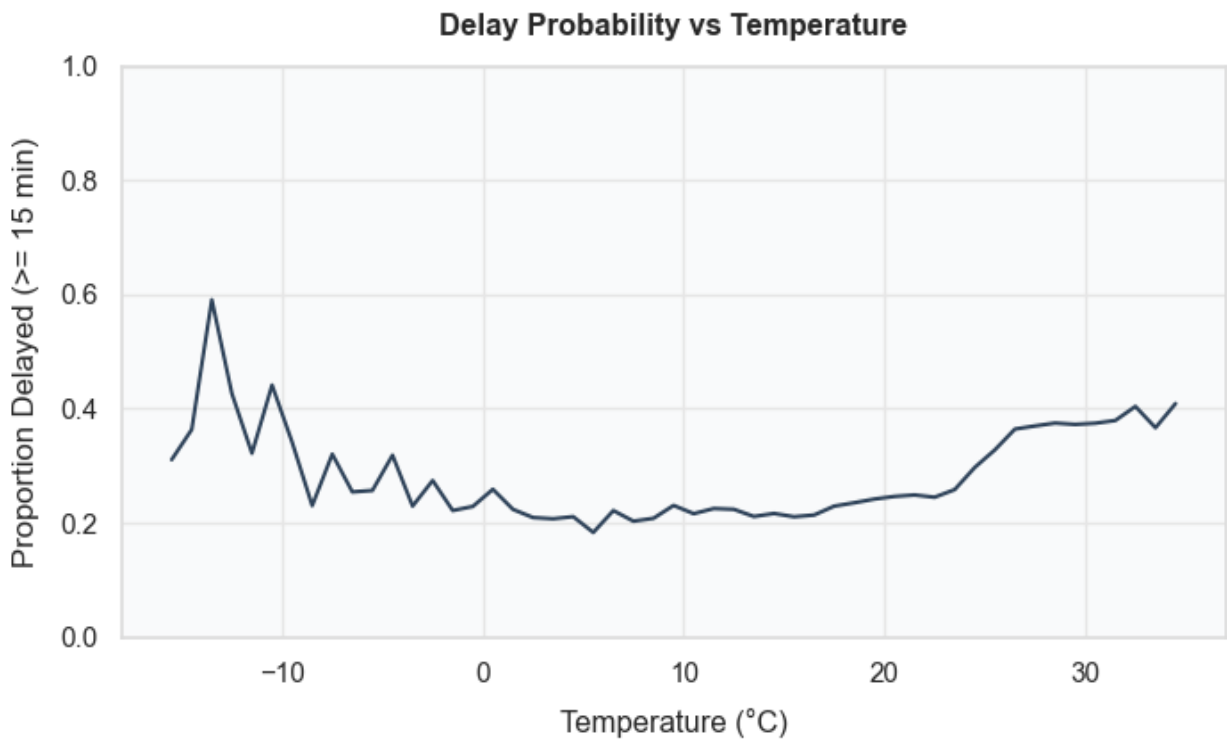
Train data exploration, analysis and visualisation

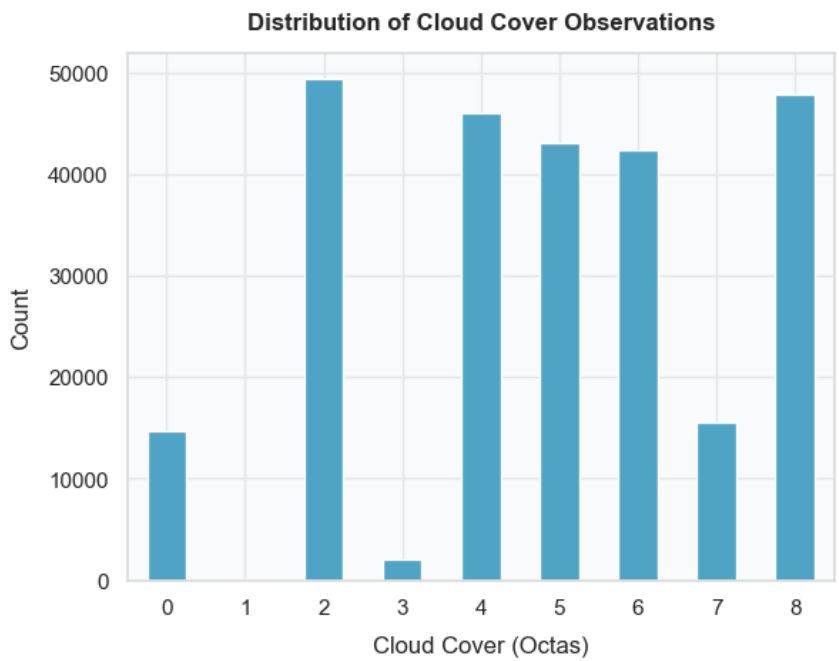
The proportion of delayed flights in the train dataset is 24.4%

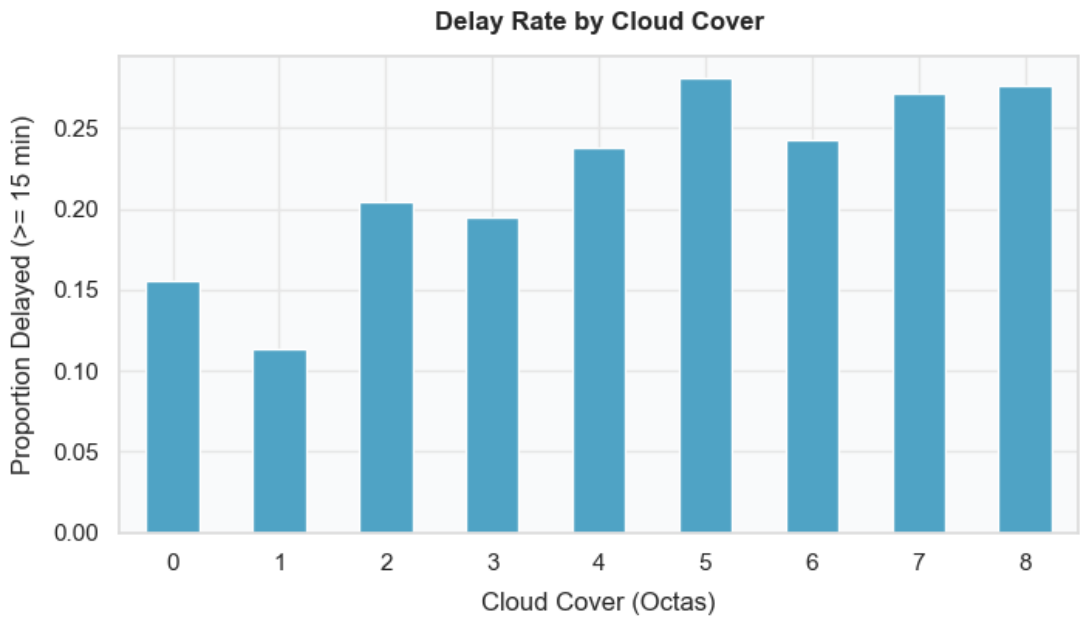


Data indicates that there could be an increased probability of flights being delayed as the temperature drops or rises.

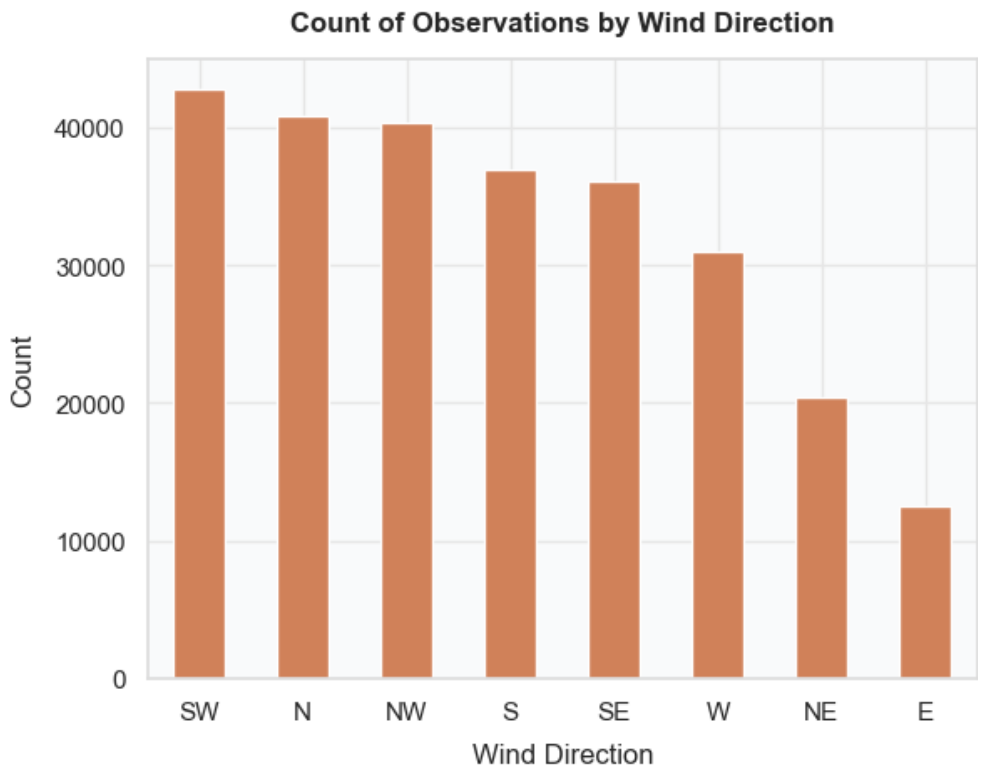
This could be associated with any de-icing procedures in cold weather or for hot weather the fact that hot air reduces the lift of an aircraft.

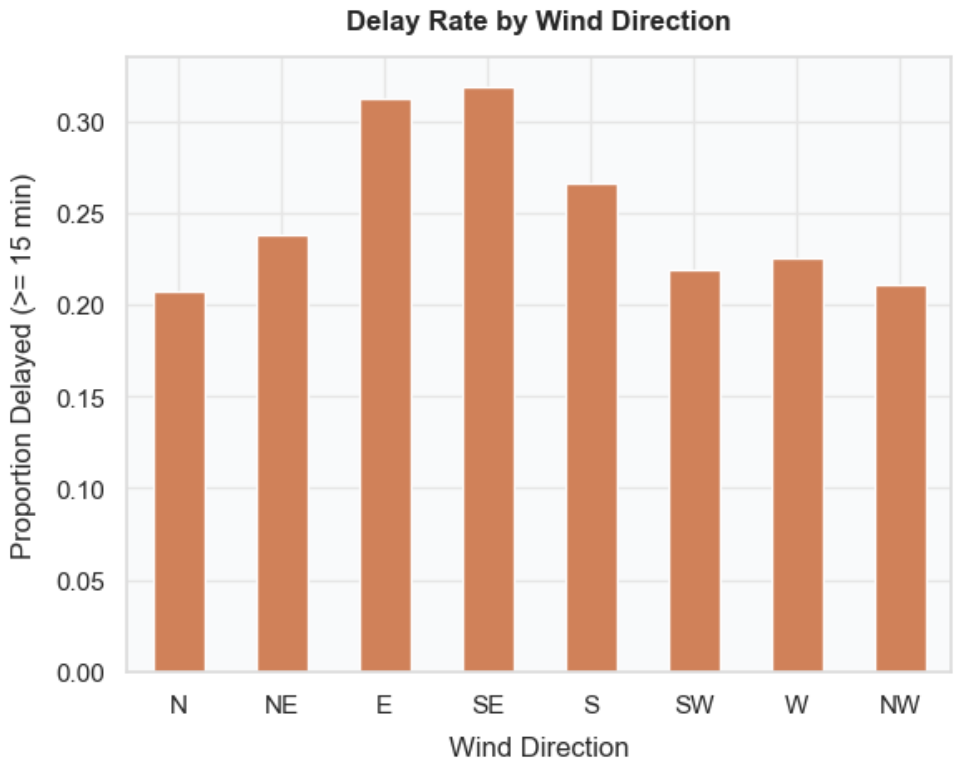




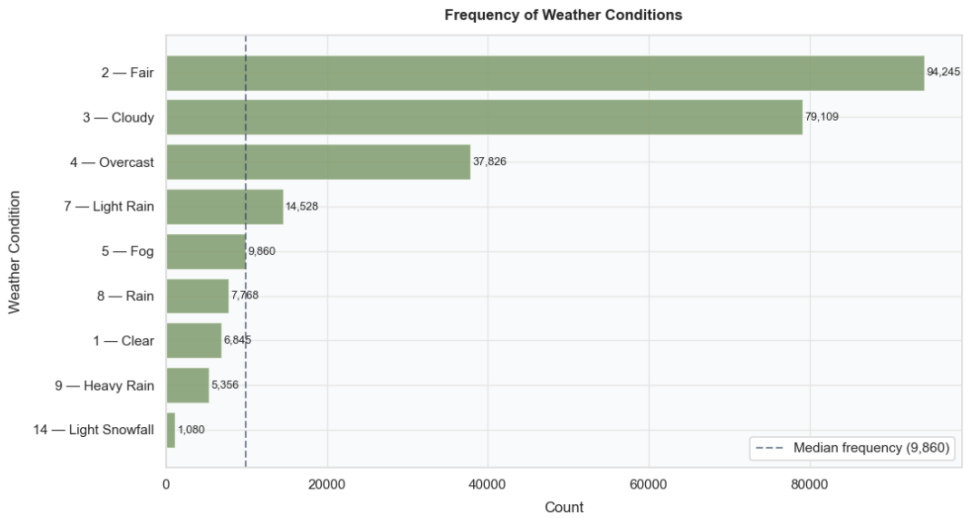


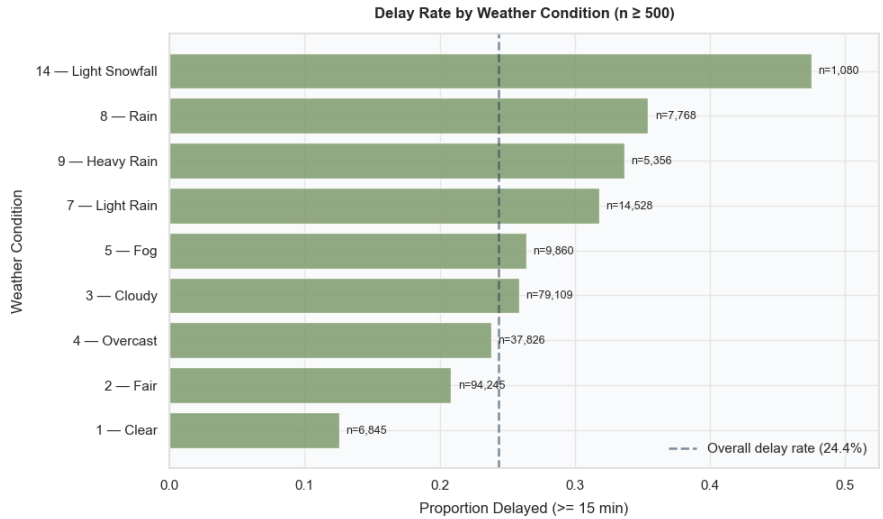
Flights departing under high cloud cover (7–8 octas) experience an approximately 10–12 percentage-point higher probability of departure delay compared to clear-sky conditions.





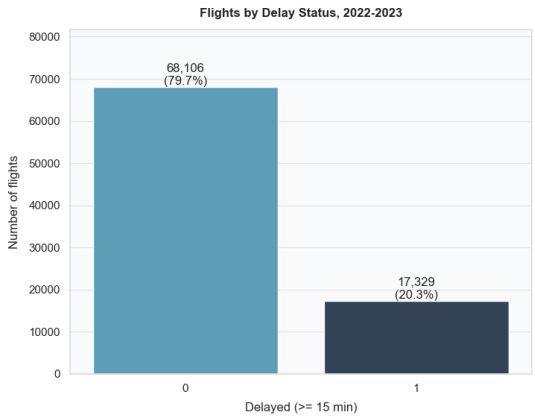
Probability of departure delay increases with the eastern and south-eastern winds.



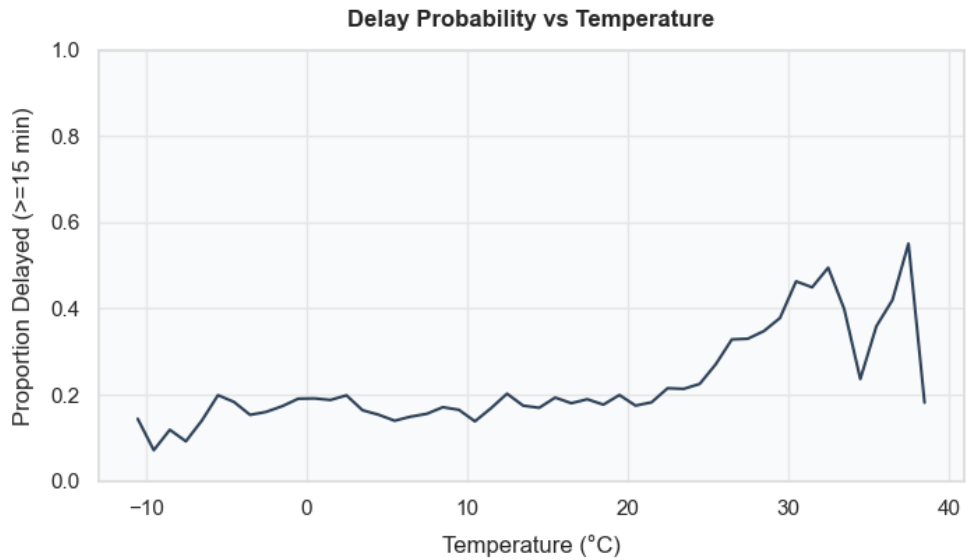


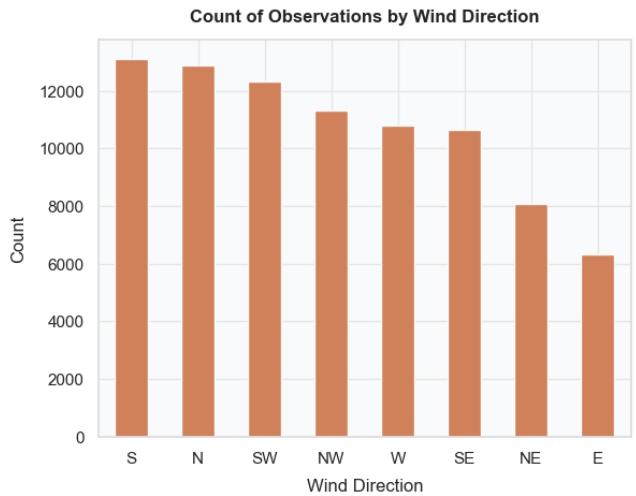
- 99% of observations range from Clear to Heavy Rain
- Flights departing under Light Snowfall condition experience an approximately 12-13 percentage-point higher probability of departure delay compared to other weather conditions.

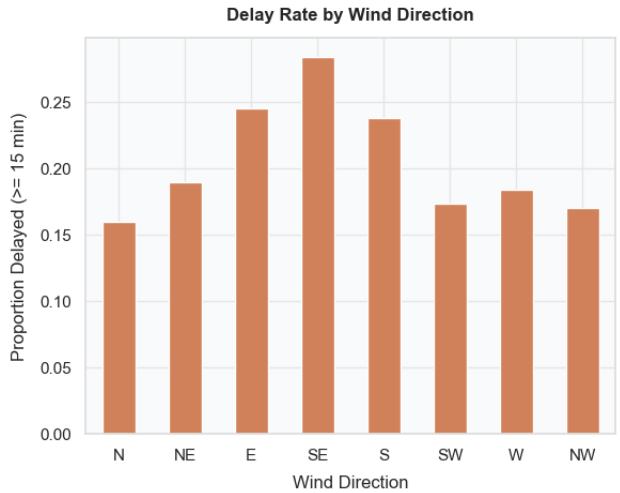
Test data exploration

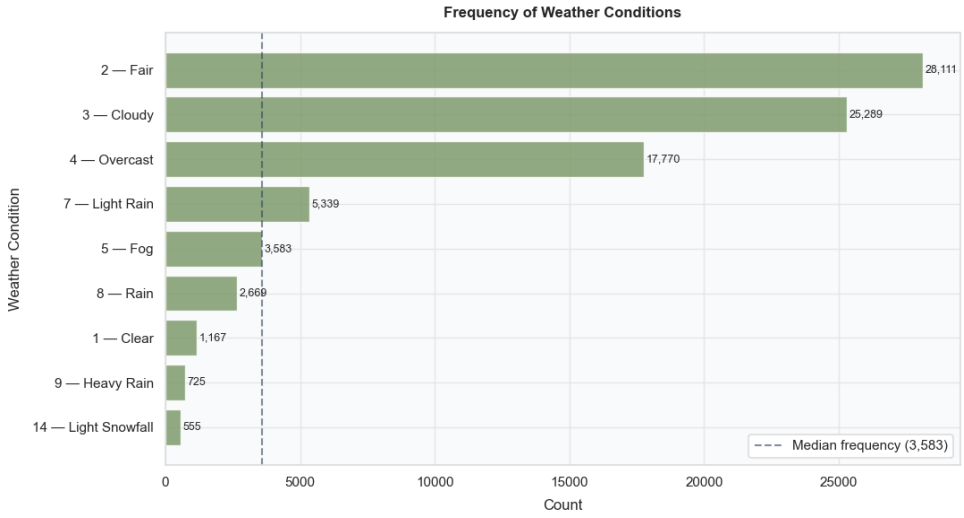


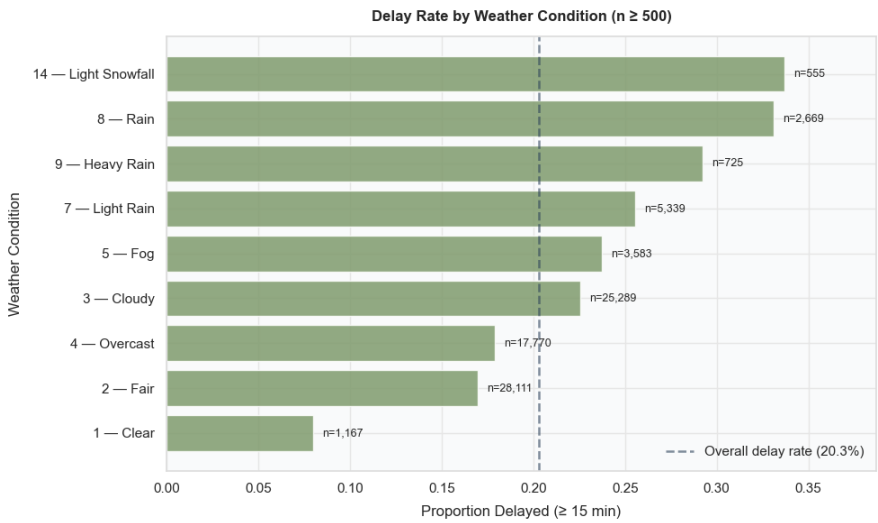
- Delayed flights rate is 20.3%
- Increased probability of delays in hot weather
- increased probability of delays with eastern and south-eastern winds









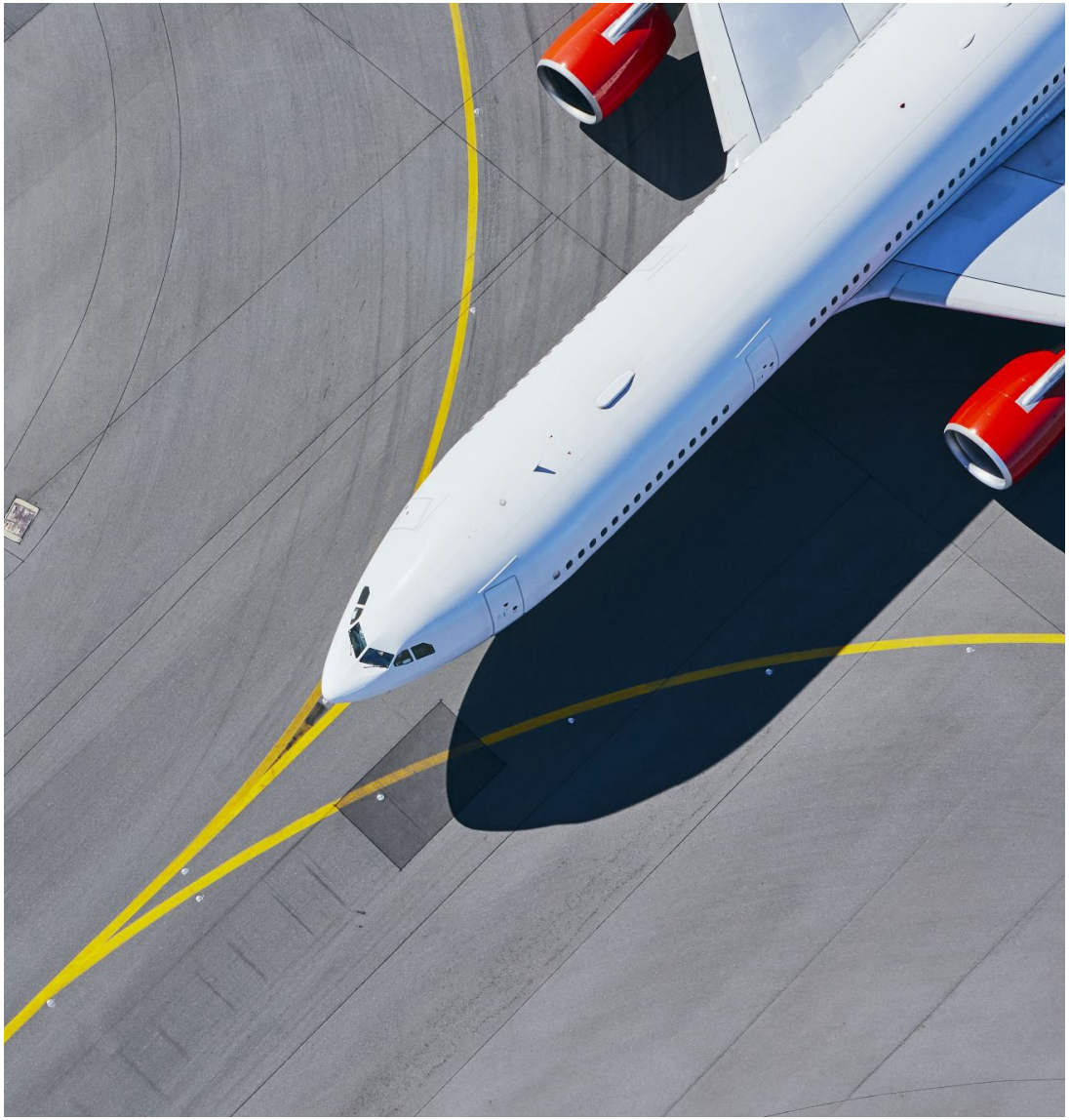


- 99% of observations for the weather conditions code range from Clear to Heavy Rain.
- With the remaining 1%, 70% of observations are Light Snowfall which again shows increased probability of flight delays.

Models used

- Logistic Regression – scaled data
- Decision Tree
- Random Forest
- XGBoost

Logistic Regression, Decision Tree, and Random Forest were all optimised with GridSearchCV



Results evaluation

Model	Recall_Pos	PR_AUC	Balanced_Acc
Random Forest	0.60	0.32	0.60
XGBoost	0.55	0.30	0.59
Logistic Regression	0.65	0.29	0.59
Decision Tree	0.54	0.27	0.56

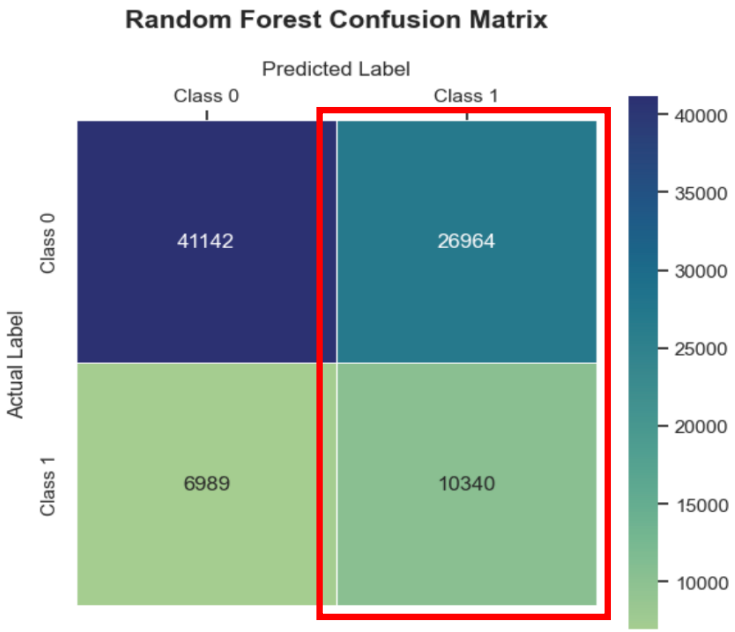
Model evaluation prioritised recall for delayed departures alongside Precision–Recall AUC to assess overall performance on the minority class.

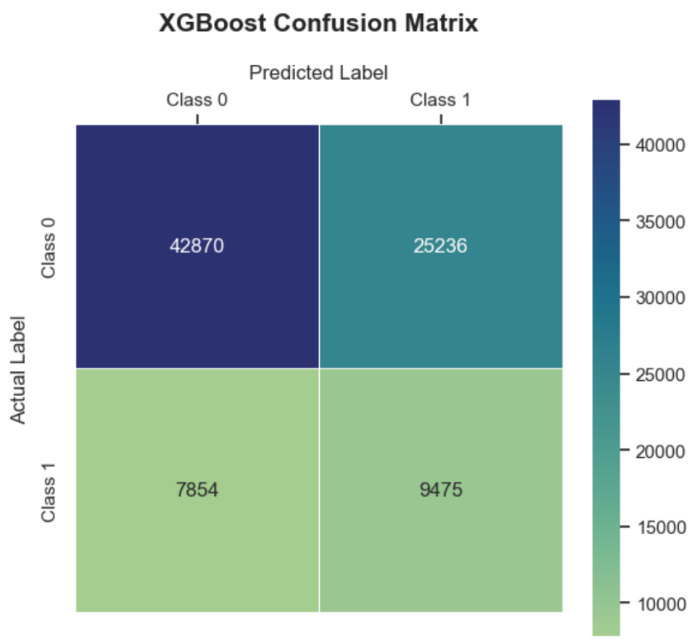
Random Forest delivered the strongest balance between detection capability and precision, achieving a recall of 0.60 and a PR AUC of 0.32.

It also achieved the highest balanced accuracy, indicating consistent performance across both delayed and on-time flights.

While this does not capture every event, many flight disruptions stem from factors beyond weather — such as inbound aircraft delays or carrier-related constraints — which were outside the scope of this model

Results evaluation



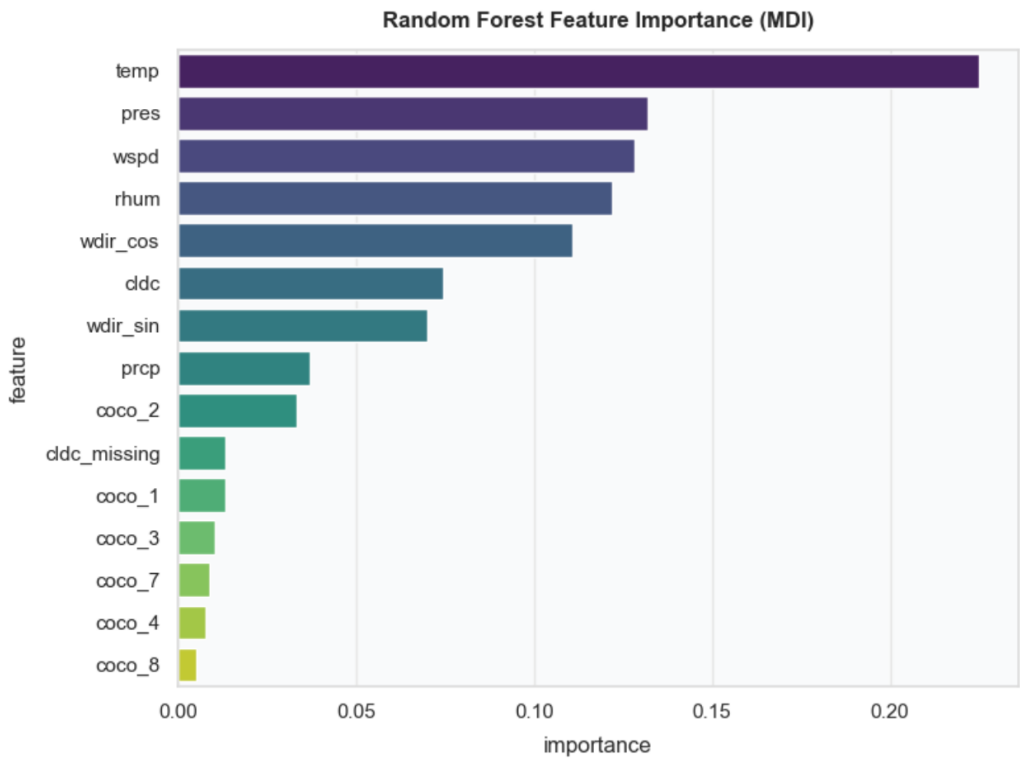


Random Forest improves delay detection by roughly 9% relative to XGBoost, while increasing false positives by about 7%.

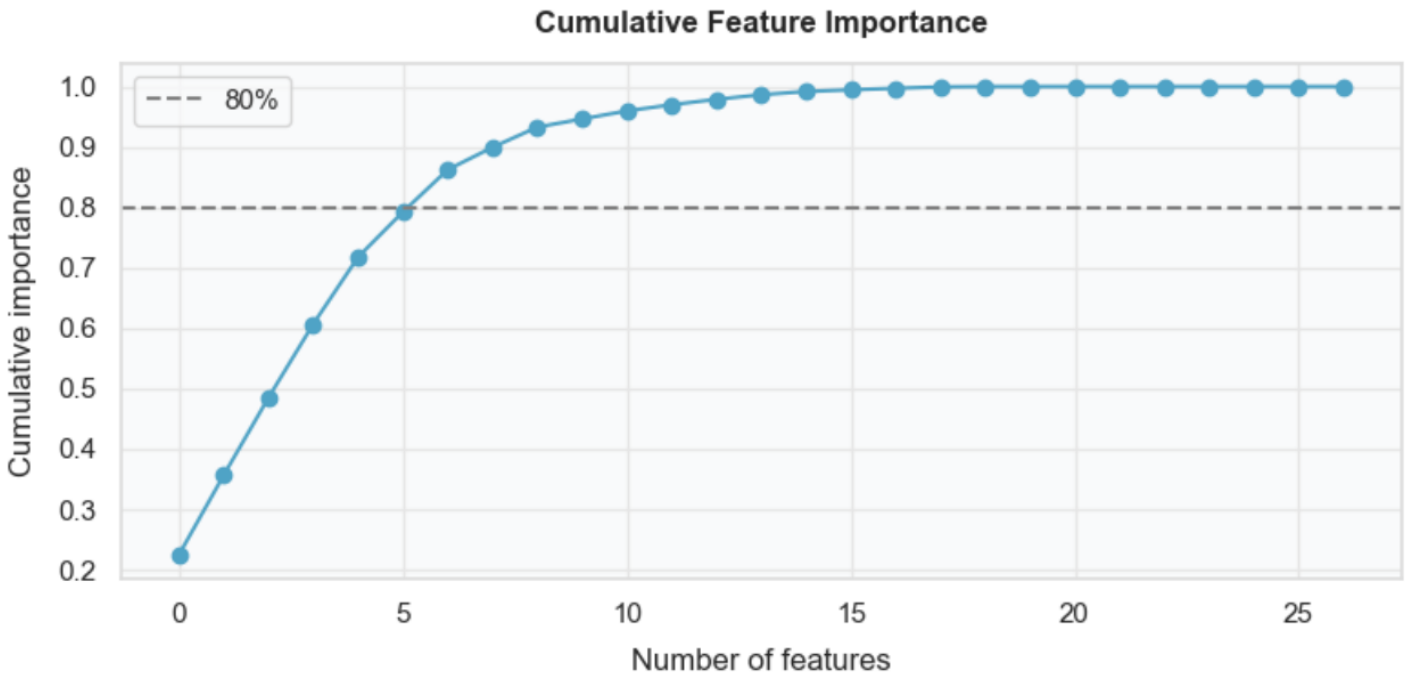
Given that missed delays typically carry greater operational risk than unnecessary alerts, this represents a favourable trade-off.

Feature Importance

Temperature, pressure, wind speed, wind direction and humidity are the primary drivers of predicted delay risk. That combination often signals storm systems.



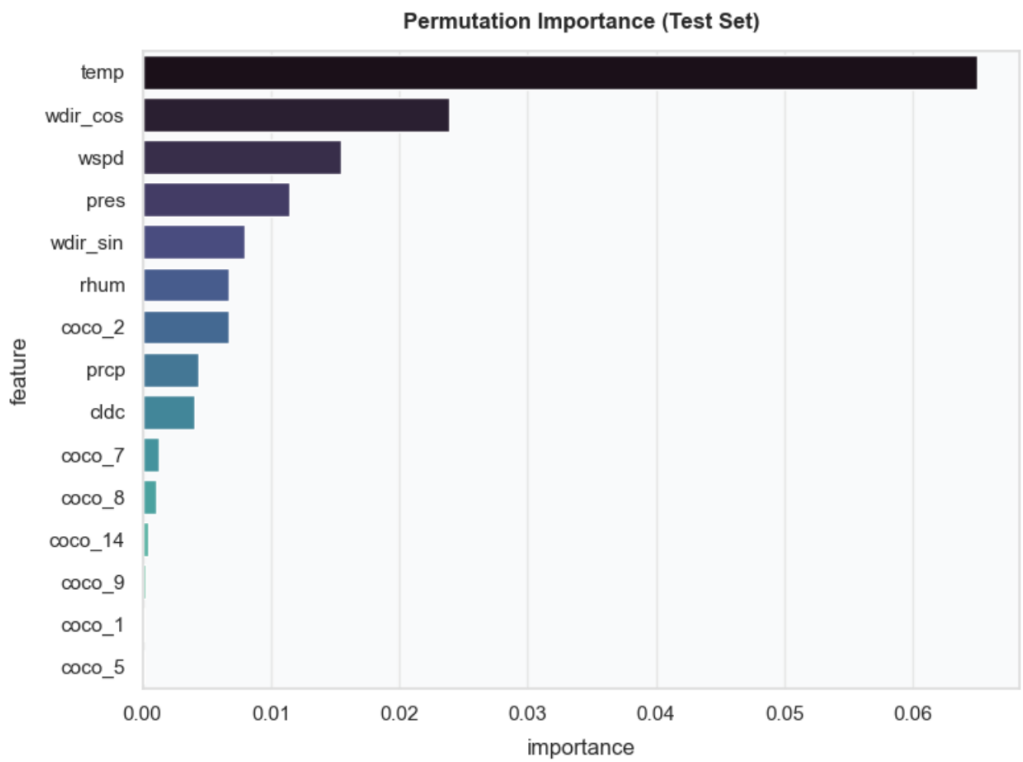
Feature Importance



Approx. 80% of total importance is captured by 5 features. After 10 features, gains are marginal.

Feature Importance

A small set of continuous meteorological variables accounts for most of the model's predictive power, with temperature contributing the largest marginal effect. Since temperature often acts as a seasonal proxy, the model is sensing weather regimes, not isolated variables.



Summary, conclusions and next steps

- The objective was to develop a predictive model using combined flight and meteorological data
- We explored several machine learning models
- The Random Forest model proved to offer best balance between maximising the number of predicted delays and false positives, and we recommend this model given the industry specifics
- Final output could be a dashboard with upcoming flights information (e.g. 3-7 days out), with meteorological data merged in, and an indicator on which flights are at a risk of being delayed
- For next steps, we recommend:
  - Testing the model on other months, perhaps with a higher ratio of delayed flights
  - Testing the model on other origins of departure to see how it holds
  - Incorporating other predictors such as days of week, month, carrier, etc.

Thank you!

Any questions?

Appendices

Random Forest

```
Classification Report (test):
              precision    recall  f1-score   support

           0       0.85      0.60      0.71     68106
           1       0.28      0.60      0.38     17329

    accuracy                           0.60     85435
   macro avg       0.57      0.60      0.54     85435
weighted avg       0.74      0.60      0.64     85435
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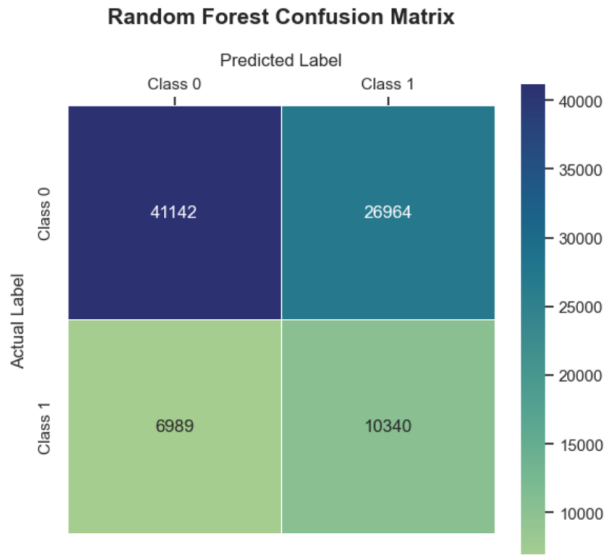

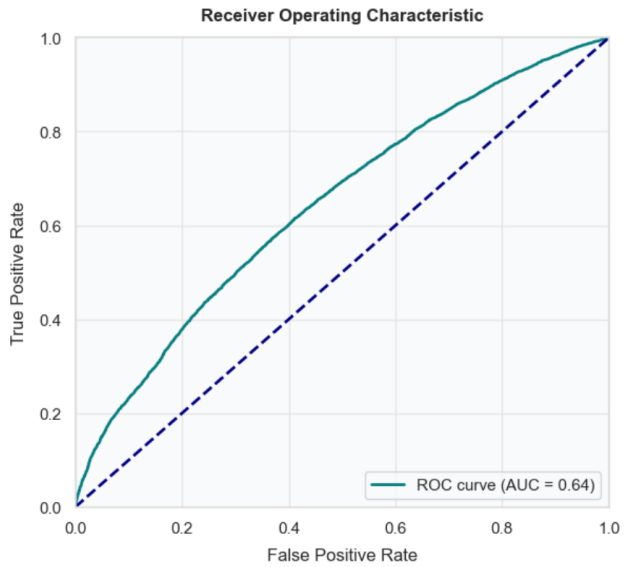

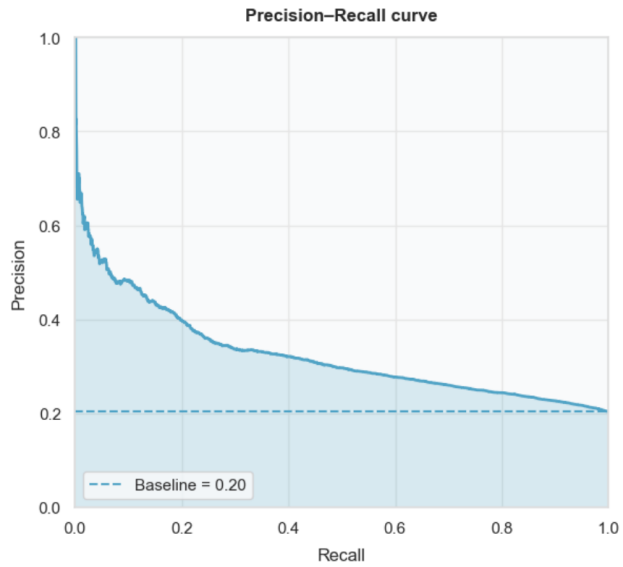

All models, comparison							
Model	Recall_Pos	PR_AUC	Balanced_Acc	TP	FP	TN	FN
Random Forest	0.60	0.32	0.60	10340	26964	41142	6989
XGBoost	0.55	0.30	0.59	9475	25236	42870	7854
Logistic Regression	0.65	0.29	0.59	11228	32354	35752	6101
Decision Tree	0.54	0.27	0.56	9416	28214	39892	7913